

E-Commerce Customer Churn Analysis

SQL Data Cleaning, Transformation & Analysis Report

Tools Used: MySQL, MySQL Workbench, SQL

Academic Project Report

1. Introduction

Customer churn is an important concern in the e-commerce industry, as businesses need to understand why customers stop engaging with their services. This project focuses on analyzing customer churn using historical e-commerce customer data.

The dataset contains information related to customer tenure, preferred payment modes, satisfaction scores, purchase behavior, device usage, order categories, complaints, and other customer attributes. SQL is used to clean, transform, explore, and analyze the dataset.

The main purpose of this project is to identify patterns related to customer churn and understand customer behavior through data analysis.

2. Problem Statement

In the e-commerce domain, businesses face the challenge of understanding customer churn patterns to maintain customer satisfaction and sustained profitability.

This project analyzes historical customer data to investigate the factors and patterns associated with customer attrition. Customer attributes such as tenure, preferred payment modes, satisfaction scores, and purchase behavior are analyzed to understand customer churn.

The objective is to provide useful insights that can help e-commerce businesses understand customer behavior and support customer retention strategies.

3. Objectives

- To clean and prepare the e-commerce customer dataset.
- To handle missing values and outliers in the dataset.
- To standardize inconsistent categorical values.
- To transform the dataset by renaming columns and creating new columns.
- To analyze churned and active customers.
- To investigate customer behavior based on payment mode, order category, satisfaction, coupons, and purchase behavior.
- To analyze the relationship between warehouse distance and customer churn.
- To analyze return information of customers who have churned and made complaints.

4. Dataset

The project uses an **E-Commerce Customer Churn** dataset containing customer-level information. The dataset includes attributes related to customer ID, churn status, tenure, preferred login device, city tier, warehouse-to-home distance, preferred payment mode, gender, hours spent on app, registered devices, preferred order category, satisfaction score, marital status, number of addresses, complaint status, order amount hike, coupon usage, order count, days since last order, and cashback amount.

The original dataset contains the columns **Churn** and **Complain**, which are transformed into more descriptive fields during the project.

5. Data Cleaning

5.1 Handling Missing Values

Mean imputation was performed for **WarehouseToHome**, **HourSpendOnApp**, **OrderAmountHikeFromlastYear**, and **DaySinceLastOrder**. The calculated mean values were rounded to the nearest integer where required.

Mode imputation was performed for **Tenure**, **CouponUsed**, and **OrderCount**. These steps were performed to handle missing values while retaining the available customer records.

5.2 Handling Outliers

Rows where the value of **WarehouseToHome** was greater than **100** were deleted as specified in the assignment.

5.3 Handling Inconsistencies

Original Value	Standardized Value
Phone	Mobile Phone
Mobile	Mobile Phone
COD	Cash on Delivery
CC	Credit Card

This ensures consistency among categorical values.

6. Data Transformation

6.1 Column Renaming

Original Column	New Column
PreferedOrderCat	PreferredOrderCat
HourSpendOnApp	HoursSpentOnApp

6.2 Creating New Columns

ComplaintReceived

Complain = 1 → Yes; otherwise → No

ChurnStatus

Churn = 1 → Churned; otherwise → Active

6.3 Dropping Columns

After creating the new columns, the original columns **Churn** and **Complain** were removed. These transformations follow the assignment requirements.

7. Data Exploration and Analysis

The cleaned and transformed dataset was analyzed using SQL. The analysis included:

- Count of Churned and Active customers.
- Average tenure and total cashback of churned customers.
- Percentage of churned customers who complained.
- City tier with the highest number of churned Laptop & Accessory customers.
- Most preferred payment mode among active customers.
- Total order amount hike for single customers preferring mobile phones.
- Average registered devices among UPI users.
- City tier with the highest number of customers.
- Gender with the highest coupon usage.
- Customer count and maximum app usage hours for each preferred order category.
- Total order count for Credit Card users having the maximum satisfaction score.
- Average satisfaction score of customers who complained.
- Preferred order categories among customers who used more than five coupons.
- Top three preferred order categories based on average cashback.
- Preferred payment modes of customers with tenure of 10 months and more than 500 orders.
- Distance-based customer categorization and churn-status breakdown.
- Order details of married customers living in City Tier 1 whose order count is above the overall average.

8. Customer Returns Analysis

A separate table named **customer_returns** was created in the **ecomm** database. The table contains **ReturnID**, **CustomerID**, **ReturnDate**, and **RefundAmount**.

Eight return records were inserted as specified in the assignment.

A final SQL **INNER JOIN** was performed between **customer_returns** and **customer_churn** using **CustomerID**. The final result displays return details along with customer details for customers who have **ChurnStatus = Churned** and **ComplaintReceived = Yes**.

9. Tools and Technologies Used

- MySQL

- MySQL Workbench
- SQL

SQL commands were used for data cleaning, transformation, analysis, table creation, insertion, and joining.

10. Conclusion

The **E-Commerce Customer Churn Analysis** project demonstrates how SQL can be used to clean, transform, and analyze customer data.

The project involved handling missing values, removing specified outliers, standardizing inconsistent values, creating meaningful customer status columns, and performing multiple analytical queries.

The analysis provides information about customer churn, payment preferences, order behavior, satisfaction, coupon usage, warehouse distance, cashback, and customer returns. The resulting information can help understand customer behavior and support data-driven customer retention strategies.

11. Screenshots / SQL Output Evidence

Insert screenshots of the SQL queries and their corresponding output results in this section as required by the assignment.